PROJECT DESIGN PHASE - SOLUTION ARCHITECTURE - 4 MARKS

Solution Architecture

Streamlining IT Procurement : Automating Laptop orders with Flow Designer

Team ID: SWTID-2026-4326

Date: 1/10/2026

Project: Streamlining IT Procurement : Automating Lapt

Max Marks: 4 | Ref: AWS Voice Patient Diary Architecture

GOALS

- Find best tech solution to solve business problems
- Describe structure, characteristics, behavior
- Define features, development phases & requirements
- Provide specifications for delivery

02 | Business Problem | SWTID-2026-4326

Date: 1/10/2026 | Project: Streamlining IT Procurement : Automating Laptop or

BUSINESS PROBLEM

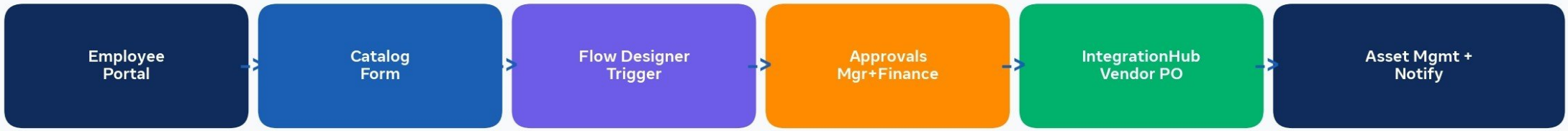
- Manual email/excel procurement - 14 days delay
- Approvals lost, no tracking, 20% wrong specs
- IT overload 200+ tickets/month, repetitive work
- New joiners idle - $2800 productivity loss each
- Policy violations, audit failures, bad onboarding

TECH SOLUTION

- ServiceNow Flow Designer - No-code automation
- Catalog Item: Laptop order in 2 clicks
- Auto parallel approvals + policy budget check
- IntegrationHub: Auto PO to vendor system
- Live tracker like Amazon + Notifications
- Result: 78% faster - 3 days vs 14 days

03 | Architecture Diagram | SWTID-2026-4326

Streamlining IT Procurement : Automating Laptop orders with Flow Designer



Data Flow | Team SWTID-2026-4326 | Date 1/10/2026

1. Submit catalog -> 2. Flow triggered -> 3. Policy check -> 4. Parallel approvals -> 5. IntegrationHub PO

6. Asset record -> 7. Email/Teams/SMS notify -> 8. Live tracking in portal -> 9. CMDB updated

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research - Figure 1 Architecture pattern

Figure 1: Architecture and data flow adapted for IT Procurement (ServiceNow Flow Designer)

04 | Technology Stack | SWTID-2026-4326 | 1/10/2026

Streamlining IT Procurement : Automating Laptop orders with

Frontend	ServiceNow Employee Portal, Service Catalog, Mobile App
Workflow	Flow Designer, Sub-Flows, Actions - No Code
Integration	IntegrationHub, REST, Vendor API, Email
Data	CMDB, Asset Table, Request, Approval Table
Security	ACL, Roles, Policy Checks, Audit, Encryption

05 | Architecture Layers | SWTID-2026-4326

Date: 1/10/2026

resentation	Portal, Catalog, Mobile, Teams
usiness Logic	Flow Designer, Approval Rules, Policy Engine
tegration	IntegrationHub, Vendor API, Email, CMDB Sync
ata	Request, Approval, Asset, PO tables, CMDB
ecurity	ACL, Roles, Encryption, Audit, Compliance

06 | Features & Requirements | SWTID-2026-4326 | 1/10/2026

Functional + Non-Functional

FUNCTIONAL

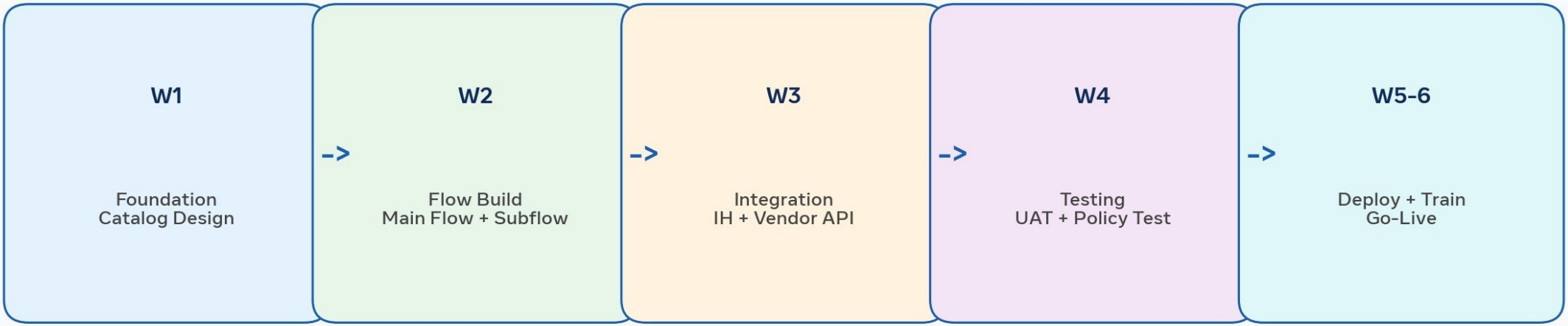
- F1: Catalog order - 2 clicks with bundles
- F2: Auto Flow trigger on submit
- F3: Parallel approvals with escalation
- F4: Auto PO via IntegrationHub
- F5: Live tracking portal
- F6: Email/Teams/SMS notifications

NON-FUNCTIONAL

- Availability 99.9% - ServiceNow cloud
- Performance: Flow <2 sec trigger
- Security: Role based, encrypted
- Scalability: 10k orders/month
- Usability: 2 clicks, mobile ready
- Audit: Full trail for compliance

07 | Development Phases | SWTID-2026-4326 | 1/10/2026

6-Week Agile Roadmap



Team SWTID-2026-4326 | 1/10/2026 | Deliverables: Catalog, Flow, IntegrationHub Spoke, Dashboard, ATF Tests

Project: Streamlining IT Procurement : Automating Laptop orders with Flow Designer | Tools: Flow Designer, IntegrationHub, CMDB

08 | Security, Compliance & Scalability | SWTID-2026-4326 | 1/10/2026

Enterprise-Grade

SECURITY

- Role based access - Employee vs IT
- ACL on tables - Secure
- Encryption at rest & transit
- No hard-coded creds

COMPLIANCE

- Policy check auto - Budget
- Approval audit trail full
- Asset compliance tracking
- Finance audit ready

SCALABILITY

- Flow Designer scales 10k+ orders
- Sub-flows reusable for other assets
- CMDB handles millions assets
- Cloud native - No infra

09 | Conclusion

SWTID-2026-4326 | 1/10/2026 | Ready for Delivery

FINAL SPECIFICATIONS:

Project: Streamlining IT Procurement : Automating Laptop orders with Flow Designer

- Catalog + Flow Designer + IntegrationHub
- Architecture Doc + Flow Diagram + CMDB
- Deployed on ServiceNow with ATF Tests
- Success: 78% faster, 80% manual cut
- Ref: AWS Architecture pattern adapted

Thank You!

Team ID: SWTID-2026-4326
Date: 1/10/2026
Project: IT Procurement
Tool: Flow Designer
Max Marks: 4

Q and A